Elias Fink

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Education

University of Cambridge
PhD in Theoretical Physics 2026 - 2030

- Research on Black Holes in the context of the Gauge-Gravity Duality and problems motivated by Quantum Gravity
- Supervised by Jorge Santos

MASt Part III of the Mathematical Tripos (Theoretical Physics) 2025 - 2026

- Pass with Distinction
- Examined in areas of General Relativity, Quantum Field Theory and Statistical Field Theory
- Part III Essay on "Perturbations of Higher Dimensional Black Holes" supervised by Jorge Santos

Imperial College London
BSc Physics with Theoretical Physics 2022 - 2025

- Graduated top of the year in final examinations
- First-class honours
- BSc project on "Quiver Gauge Theories" with Prof Amihay Hanany

Rossall School, Lancashire
International Baccalaureate (IB) Diploma with total: 42 2020 - 2022

Technical Experience

Cosmology Research Internship at Abdus Salam Centre for Theoretical Physics June - September 2025

- Worked with Prof Carlo Contaldi and PhD student Giorgio Mentasti to compute two-point correlation functions for a Stochastic Gravitational Wave Background
- Cross-correlated short-distance and long-distance limits to determine new results for astrometric measurements and pulsar timing arrays
- Preprint available at arXiv:2510.25646

Inertial Confinement Fusion Research Internship at York Plasma Institute July - August 2024

- Automated analysis of RCF stacks for proton radiography to find integrated magnetic fields in Inertial Confinement Fusion plasma
- Created software to design phase plates to smooth out or customise laser intensity profile
- Simulated proton radiography that would be generated from Biermann Battery magnetic field

Further Experience

College Exam Invigilator at St Edmund's College, Cambridge December 2025 - present

- Invigilating college examinations across various degrees
- Responsibility for upholding academic integrity and fairness

Private Online Tutoring August 2023 - present

- Tutoring at German, English and Scottish pre-university level in higher level mathematics and physics
- Advanced tutoring for students entering undergraduate physics degrees

Simulation Engineering Internship at Gratz Engineering GmbH July - August 2023

- Used FEA to analyse structures by methods of modal analysis, harmonic response, explicit dynamics, and static structural analysis for respective client firms
- Used CFD to help analyse both water cooling systems and air flow systems

Technical skills

Software skills

- Python, Swift, Java, HTML/CSS, JavaScript, C++, C#, Fortran, Mathematica, ANSYS Products

Languages

- English (native), German (native), Japanese (elementary)

Interests and Projects

Lead Organiser at the QUIRK+ Physics Conference 2026 September 2025 – February 2026

- Coordinating one of the largest student physics conferences with 70+ speakers from 15 English and European universities in collaboration with 10 physics societies
- Organising two events in January and February at King's College and the University of Cambridge
- Secured substantial funding from the IOP and Quantinuum

Directed Reading Programme Mentor December 2024 – April 2025

- Taught Quantum Mechanics and Special Relativity to Mathematics students
- Advised students for their own reading and presentations at the Department of Mathematics symposium

Third Year Research Project (high first class) October 2024 – April 2025

- Worked with Prof Amihay Hanany to analyse 3d N = 4 supersymmetric quiver gauge theories
- Computed Hilbert series of Coulomb branch and Higgs branch of moduli space of vacua and magnetic dual quivers for unitary and orthosymplectic theories that arise from brane constructions in (9+1)d type IIB superstring theory

Representative for Science to the Imperial College Union Council October 2024 - July 2025

- Attended and voted in council meetings to further the interests of Royal College of Science students at Imperial College Union (ICU)
- Proposed papers to decrease bureaucracy towards societies and make ICU operations transparent

President of the Physics Society May 2024 - July 2025

- Manage committee, events and finances for the Imperial College Physics Society (1600+ members)
- Organised large events such as the Physics Society Christmas Formal Dinner and Boat Party
- Arranged weekly guest lectures, undergraduate colloquia and laboratory tours
- Established the RCSU Research-A-thon and QUIRK London University Physics Conference

Physics UG Academic Representative October 2023 - September 2024

- Collected student feedback and alerted department staff about student experience
- Worked and collaborated across the departments and faculties to communicate the best practise
- Attended number of meetings with College and Union staff

Second Year Computing Project (high first class) May - June 2024

- Simulated gas using perfectly elastic collisions between hard spheres to extract physical parameters and confirm statistical physics such as the Maxwell-Boltzmann distribution and ideal gas law
- Improved modelling by considering virial expansion rather than ideal gas law and confirming Van der Waals equation of state with empirical virial coefficients to high accuracy

Invited Talks

- Instabilities of Higher-Dimensional Black Holes and Weak Cosmic Censorship (Imperial College, Physics Department Guest Lecture Series and Queen Mary University, PsiStar Society Lectures, March 2026)
- Quasinormal Modes of Black Strings and Myers-Perry Black Holes (Imperial College, Quantum Fields & Fundamental Forces Seminar and University of Cambridge, Part III Lent Seminar, March 2026)
- A Novel Approach for Detecting Stochastic Gravitational Waves (University of Cambridge, Part III Michaelmas Seminar, December 2025)
- Biermann Battery Magnetic Fields in Inertial Confinement Fusion (Imperial College, Undergraduate Colloquium, February 2025)
- Quiver Gauge Theories (University of Warwick, Warwick-Imperial Maths Conference, November 2024)

Academic Achievements

- Governors' BSc Prize in Physics – best performing BSc student at final examinations
- Imperial College UROP Award – funding to perform a research project for UG students
- RCSU President's Award – highest award for contributions to the Royal College of Science Union
- Stanley Raimes Memorial Prize – best performing physics student in mathematics in Y1 and Y2
- Dean's List (Year 1, Year 2, Year 3) – top 10% of the cohort